

Effect of Exhaust Gas Recirculation and Oxy-Fuel Dilution on Methane Combustion

Adiabatic Flame Temperature, Laminar Flame Speed, Auto-Ignition Delay and NO_x Formation

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Abstract

This project investigates how two carbon-management combustion strategies — *exhaust gas recirculation* (EGR) and *oxy-fuel* operation — reshape the combustion of methane relative to a conventional methane/air baseline. Four complementary quantities are computed with the open source kinetics library *CANTERA* and the GRI-Mech 3.0 reaction mechanism: the adiabatic flame temperature (constant-enthalpy equilibrium), the laminar flame speed (one-dimensional freely-propagating flame), the auto-ignition delay (zero-dimensional constant-pressure reactor) and the equilibrium NO concentration. The results quantify the well-known trade-off behind low-emission combustor design: both EGR and CO_2 -diluted oxy-fuel combustion lower the flame temperature, slow the flame and lengthen the ignition delay, but in exchange they suppress thermal NO by up to an order of magnitude — and oxy-fuel combustion, by eliminating atmospheric nitrogen, removes thermal NO almost entirely while yielding a CO_2 -rich exhaust suited to carbon capture.

1 Introduction

Stationary gas turbines and reciprocating engines burning natural gas are under growing pressure to cut both nitrogen-oxide (NO_x) emissions and the carbon-capture penalty of their exhaust. Two strategies dominate the discussion. In **exhaust gas recirculation** (EGR) a fraction of the cooled burned gas is re-inducted with the fresh charge; the additional CO_2 and H_2O act as a thermal ballast that lowers the peak temperature and, through the strong temperature sensitivity of the Zeldovich mechanism, throttles thermal NO . In **oxy-fuel** combustion the nitrogen of air is replaced by recycled CO_2 , so the fuel burns in an O_2/CO_2 atmosphere; this both removes the nitrogen that feeds thermal NO and produces an exhaust that is essentially CO_2 and water vapour, from which CO_2 can be separated by simple condensation.

Both strategies, however, degrade combustion: a colder flame is slower, harder to ignite and more prone to instability. The purpose of this project is to make that trade-off quantitative for methane, the principal constituent of natural gas, using a single consistent chemical mechanism so that the temperature, flame-speed, ignition and emission effects can be compared on equal footing.

2 Literature review

The adiabatic flame temperature and equilibrium composition of hydrocarbon/air mixtures are classical results of chemical thermodynamics and are reproduced here only as a baseline. The laminar burning velocity of stoichiometric methane/air at ambient conditions is firmly

established at 0.36–0.40 m/s both experimentally and with detailed mechanisms such as GRI-Mech 3.0, and peaks slightly rich, near $\phi \approx 1.05$ –1.10. The temperature dependence of methane auto-ignition follows Arrhenius behaviour with an apparent activation energy of roughly 2×10^5 J/mol, spanning several orders of magnitude across the 1000–1800 K window relevant to engines.

The suppression of thermal NO by flue-gas dilution is the physical basis of practically every low-NO_x gas-turbine combustor, and oxy-fuel combustion with flue-gas recirculation is a leading candidate for carbon-capture-ready power generation. The replacement of N₂ by CO₂ is known to reduce the burning velocity markedly, because CO₂ has a substantially higher molar heat capacity than N₂ and because it participates chemically through $\text{CO}_2 + \text{H} \rightleftharpoons \text{CO} + \text{OH}$, which competes with the chain-branching pool. The present study reproduces these established trends and collects them into a single comparative framework.

3 Numerical model

3.1 Software and chemistry

All computations use CANTERA 3.2 with the GRI-Mech 3.0 mechanism (53 species, 325 reactions), which was developed and validated for natural-gas combustion and includes the nitrogen sub-mechanism required for NO. Unless stated otherwise the unburned/reference state is $T_u = 300\text{K}$ and $p = 1\text{atm}$.

3.2 Mixture definitions

The fuel is methane and the stoichiometric reference is $\text{CH}_4 + 2(\text{O}_2 + 3.76\text{N}_2)$. The equivalence ratio ϕ scales the fuel relative to this stoichiometric oxidiser. Three families of fresh charge are studied:

- **Air** — CH_4 with air ($\text{O}_2:\text{N}_2 = 1:3.76$).
- **Oxy-fuel** — the oxidiser is an O_2/CO_2 blend with a prescribed O_2 mole fraction x_{O_2} ; $x_{\text{O}_2} = 0.21$ reproduces the oxygen content of air but with CO_2 as the bath gas.
- **EGR** — a mole fraction f of cooled burned gas is mixed back into the fresh CH_4/air charge. The recirculated gas is taken from the equilibrium products at the same ϕ , restricted to the stable bulk species (CO_2 , H_2O , N_2 , O_2 , CO); the trace radicals present at flame temperature are assumed to recombine during cooling before re-induction.

3.3 Governing models

Adiabatic flame temperature. The burned state is obtained from a constant-enthalpy, constant-pressure (HP) equilibrium of the fresh mixture,

$$h(T_{ad}, p, Y_{eq}) = h(T_u, p, Y_u), \quad \mu_i \text{ minimised at fixed } (h, p). \quad (1)$$

Laminar flame speed. A one-dimensional freely-propagating premixed flame is solved for the conservation of mass, species and energy with mixture-averaged transport. The eigenvalue of the problem is the mass burning rate; the laminar flame speed is $S_L = \dot{m}/\rho_u$, evaluated at the cold boundary. The grid is adaptively refined until the solution is grid-independent to within a few percent.

Auto-ignition delay. A zero-dimensional adiabatic, constant-pressure reactor is integrated in time from an elevated initial temperature. Ignition delay τ is defined as the instant at which the temperature has risen by 400 K above its initial value, a robust proxy for the thermal-runaway point. The reference pressure for this part is 5 atm.

Equilibrium NO. The NO mole fraction is taken from the HP-equilibrium burned gas and reported in ppm. Equilibrium NO over-predicts the absolute level reached in a finite residence time, but it preserves the correct sensitivity to flame temperature and to the availability of atmospheric nitrogen, which is the comparison of interest here.

4 Results and discussion

4.1 Adiabatic flame temperature

Figure 1a isolates the chemical effect of each diluent by adding a pure species to a stoichiometric methane/air flame. CO_2 is the most effective temperature suppressant, followed by H_2O and then N_2 : adding 28 % CO_2 lowers T_{ad} from 2225 K to about 1653 K, against 1840 K for the same amount of N_2 . The ordering follows the molar heat capacities and the endothermic dissociation of CO_2 at high temperature.

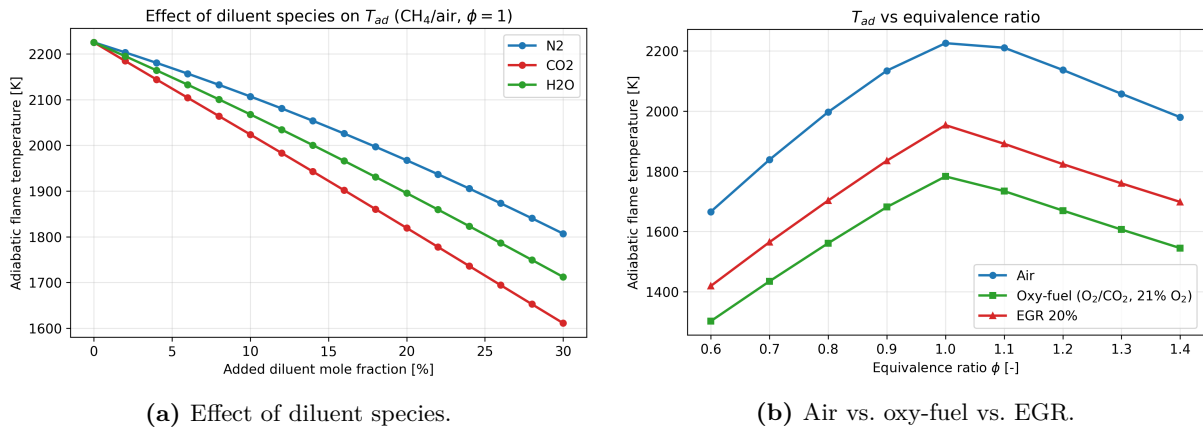


Figure 1: Adiabatic flame temperature. Left: pure-diluent study at $\phi = 1$. Right: the three combustion regimes across the equivalence-ratio range.

Figure 1b compares the three regimes across ϕ . All curves peak slightly rich, near $\phi \approx 1.0$ – 1.05 . At stoichiometry the air flame reaches 2225 K; 20 % EGR lowers this to 1957 K, and oxy-fuel at $x_{\text{O}_2} = 0.21$ falls to only 1783 K. The oxy-fuel case is the coldest precisely because its bath gas is entirely CO_2 ; in practice oxy-fuel combustors run with x_{O_2} well above 21 % to recover flame temperature, as quantified below.

4.2 Laminar flame speed

Figure 2a shows the laminar flame speed against equivalence ratio. The air baseline peaks at 39 cm/s near $\phi = 1$, in good agreement with accepted data. 10 % EGR cuts the peak to about 27 cm/s, and CO_2 -diluted oxy-fuel at $x_{\text{O}_2} = 0.30$ to about 18 cm/s. The colder, diluted flames are uniformly slower, which in a real combustor narrows the stable operating window and raises the risk of blow-off.

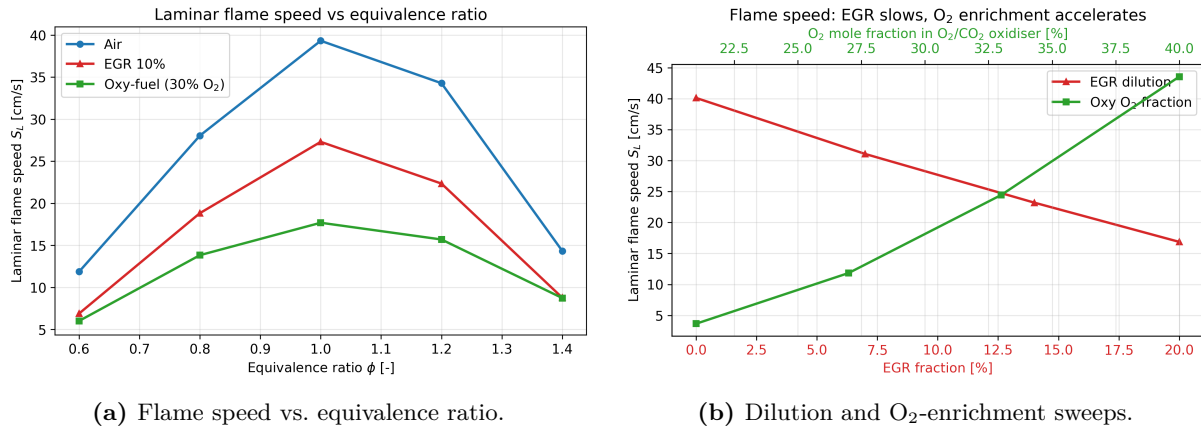


Figure 2: Laminar flame speed. Right: at $\phi = 1$, EGR monotonically slows the flame, whereas raising the O_2 fraction of the O_2/CO_2 oxidiser accelerates it.

Figure 2b sweeps the two dilution parameters at stoichiometry. Increasing EGR from 0 to 20 % drops S_L from 40 to 17 cm/s. Conversely, raising the oxygen fraction of the O_2/CO_2 oxidiser from 0.21 to 0.40 lifts S_L from a sluggish 3.7 to 44 cm/s — above the air value — which is exactly the lever used to restore flame speed and stability in practical oxy-fuel systems.

4.3 Auto-ignition delay

Figure 3a presents the ignition delay on an Arrhenius axis. All three regimes collapse onto a nearly straight line spanning four orders of magnitude between 1800 K ($\tau \approx 30\mu s$) and 1000 K ($\tau \approx 0.18s$ for air). Dilution has only a modest kinetic effect at fixed temperature, but because EGR also lowers the charge reactivity, the EGR curve sits at the longest delays on the cold (right-hand) side of the plot.

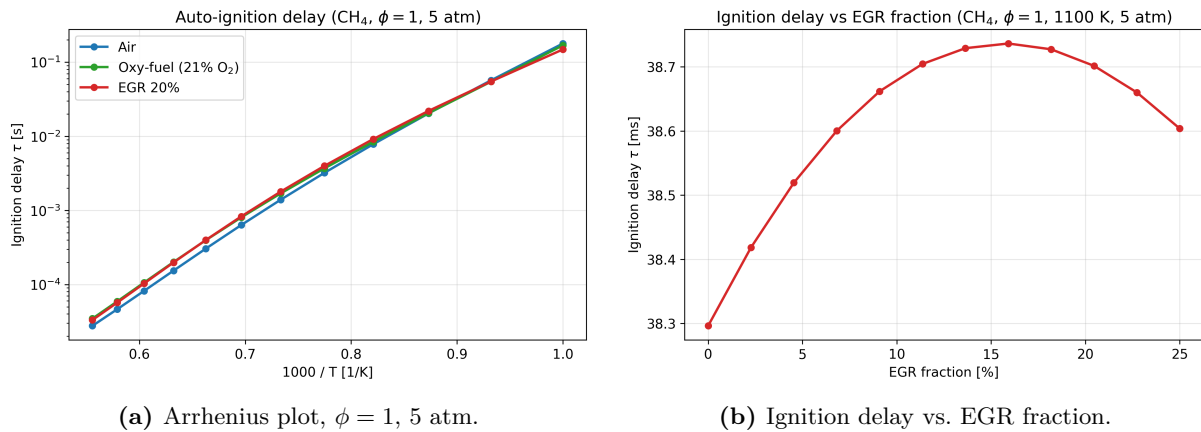


Figure 3: Auto-ignition delay in a constant-pressure reactor. EGR lengthens the delay roughly linearly with the recirculated fraction.

Figure 3b isolates the EGR effect at a fixed 1100 K: the delay grows roughly linearly with the recirculated fraction, doubling by about 25 % EGR. This lengthening is the kinetic signature of charge dilution and is one of the practical limits on how much EGR an engine can tolerate before misfire.

4.4 Pressure dependence of ignition

Figure 4 sweeps the initial pressure at a fixed 1100 K. The ignition delay follows a clean power law $\tau \propto p^{-1.09}$ across the 1–40 atm range, close to the inverse-first-order behaviour expected when

the rate-controlling chain chemistry is dominated by bimolecular reactions. Higher pressure shortens the delay sharply, which is why the engine-relevant ignition study was carried out at 5 atm rather than at atmospheric pressure.

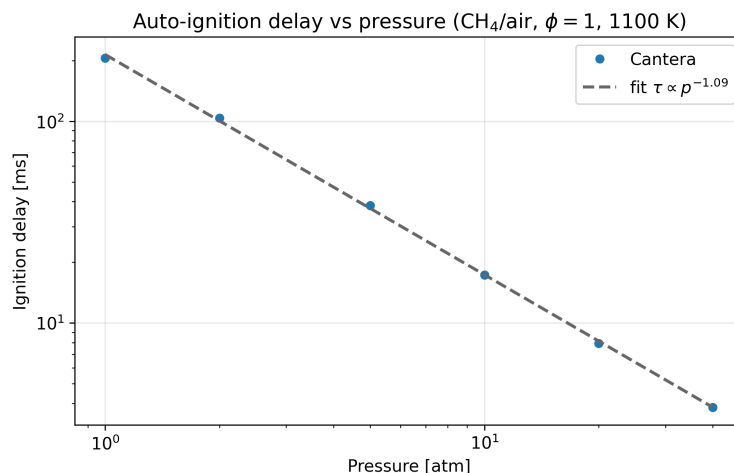


Figure 4: Auto-ignition delay versus pressure with a fitted power law.

4.5 NO_x formation

The emission payoff is summarised in Figures 5a and 5b. For air, equilibrium NO peaks at roughly 3080 ppm slightly lean ($\phi \approx 0.85$), where temperature is still high but oxygen is abundant — the classic thermal-NO maximum. 20 % EGR roughly halves the peak. Oxy-fuel combustion produces essentially *zero* NO across the whole range, simply because there is no atmospheric nitrogen to oxidise.

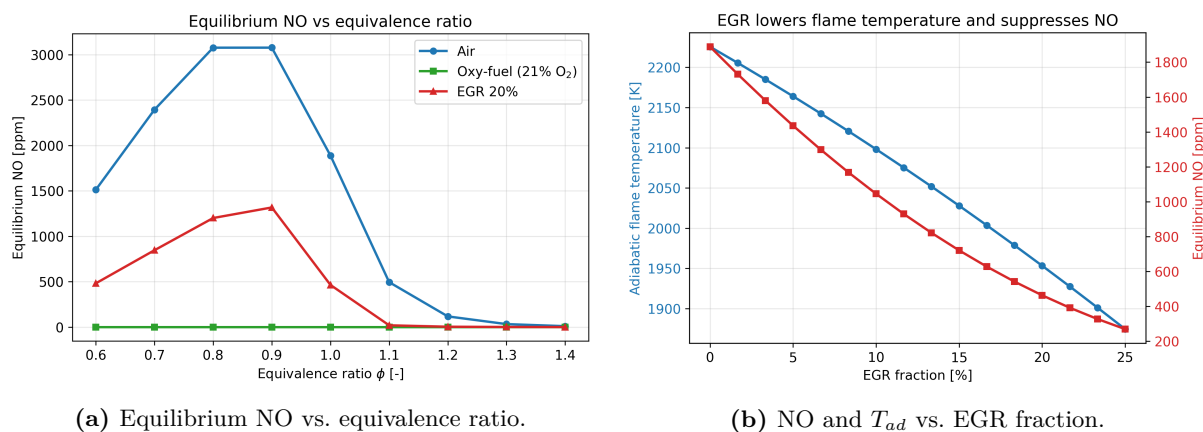


Figure 5: NO formation. The right-hand panel makes the EGR mechanism explicit: the NO reduction tracks the fall in flame temperature.

Figure 5b exposes the mechanism directly. As EGR rises from 0 to 25 %, the stoichiometric flame temperature falls from 2225 to 1880 K and equilibrium NO collapses from about 1890 to 380 ppm — a five-fold reduction driven almost entirely by the temperature drop, since thermal NO is exponentially sensitive to T . This single figure captures the central trade-off of low-emission combustion: NO is bought down with flame temperature, and flame temperature is paid for in flame speed and ignitability.

4.6 Finite-rate NO kinetics

The equilibrium NO above is an upper bound. Figure 6 shows the *kinetic* build-up of NO, obtained by integrating the full mechanism at the flame temperature with the nitrogen-oxide pool initially set to zero. Two effects combine. First, thermal NO forms slowly: even the hot air flame needs tens of milliseconds to approach its equilibrium value, so the NO actually emitted is set by the residence time. Second, and more decisive for design, the lower flame temperatures of EGR and oxy-fuel throttle the formation rate exponentially through the Arrhenius dependence of the Zeldovich mechanism — at 20% EGR the NO is still below 50 ppm after 0.5 s, two orders of magnitude under the air value, and oxy-fuel produces essentially none. The kinetic picture therefore reinforces the equilibrium result: dilution suppresses NO both thermodynamically and kinetically.

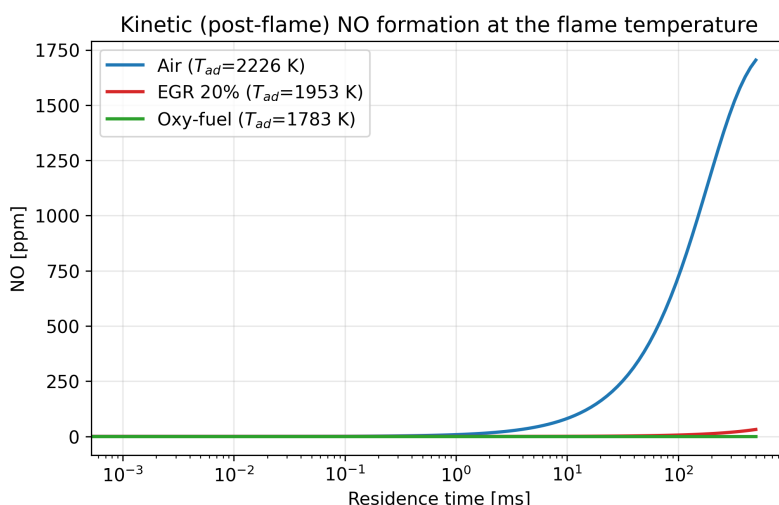


Figure 6: Kinetic (post-flame) NO formation at the flame temperature. Lower temperatures suppress NO not only at equilibrium but also kinetically.

5 Conclusions

A consistent CANTERA/GRI-Mech 3.0 framework was used to compare conventional, EGR-diluted and oxy-fuel methane combustion across four physically distinct quantities. The main findings are:

1. Both EGR and CO₂-rich oxy-fuel operation lower the adiabatic flame temperature, CO₂ being a stronger diluent than N₂ per unit mole.
2. The colder mixtures burn more slowly: 20 % EGR more than halves the laminar flame speed, while increasing the oxygen fraction of an O₂/CO₂ oxidiser is an effective way to recover — and exceed — the air flame speed.
3. Auto-ignition follows Arrhenius behaviour over four decades; EGR lengthens the ignition delay roughly linearly, setting a practical dilution limit.
4. Thermal NO falls steeply with flame temperature: EGR delivers a several-fold reduction, and oxy-fuel combustion eliminates thermal NO outright while yielding a capture-ready CO₂ exhaust. A finite-rate reactor confirms that dilution suppresses NO kinetically as well as thermodynamically, and that the emitted NO is residence-time limited.
5. Auto-ignition delay obeys a power law $\tau \propto p^{-1.09}$ in pressure, so engine-pressure operation ignites far faster than the atmospheric-pressure case.

Together the results quantify, in one place, the temperature–speed–ignition– emission trade-off that underlies modern low-NO_x and carbon-capture-ready combustor design. NO was treated at both the equilibrium and finite-rate levels; the remaining modelling limitations are the use of GRI-Mech 3.0 (validated for natural gas rather than for ammonia- or CO₂-rich oxy chemistry) and the single-reactor residence-time model. A dedicated oxy-fuel mechanism and a networked reactor model would refine the absolute emission levels without changing the qualitative conclusions.

References

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